

Section 1: Executive Summary

Write 2-3 paragraphs explaining:

- *What system you analyzed*
- *Why threat modeling matters for this system*
- *Your top 3 findings (highest priority threats)*

Write this for Marcus Webb, the General Manager. He's not technical. Use plain language.

The Grand Marina is a 500-room luxury that uses Hydroficient's Hydrologic system. They have 3 Hydrologic devices installed for the Main building, Pool/Spa and Kitchen/Laundry. The system includes cloud-based monitoring, a web dashboard, and remote shutoff capability. Hydroficient is an IoT company that makes water flow management devices for commercial properties. Recently, a \$4.2M water damage incident occurred when a leak alert was ignored. This is a security incident.

Our analysis identified three critical concerns requiring immediate attention:

- Weak authentication on the web dashboard could allow unauthorized access to flow data and alert settings,
- Unencrypted device communications could be intercepted or modified, potentially causing false readings or missed alerts,
- The central monitoring server represents a single point of failure — if compromised,

We recommend prioritizing multi-factor authentication, encrypted communications, and server redundancy.

Section 2: System Overview

Describe The Grand Marina's HYDROLOGIC system:

- *500-room luxury hotel (Hydroficient customer)*
- *3 HYDROLOGIC flow management devices (one per water service line)*
- *Cloud-based monitoring and control*

- *Web dashboard for operators and management*
- *Remote shutoff and gate control capability*

Include a simple data flow diagram (can be ASCII art or describe the flow):



Section 3: Asset Inventory

List the key assets and their CIA priorities (use your work from Step 2):

Asset	Description	C Priority	I Priority	A Priority
HYDROLOGIC Devices	3 flow management units	High	Medium	Critical
Web Dashboard	Operator monitoring interface	Critical	High	High
Cloud API	Device-to-cloud communication	High	Critical	medium
Remote Controls	Gate adjustments, emergency shutoff	Critical	High	High
Consumption Data	Savings reports, billing records	Critical	High	Medium

Section 4: STRIDE Analysis

This is the core of your threat model. For **each major component**, analyze all six STRIDE categories:

Component 1: HYDROLOGIC Devices

Threat	Scenario	Likelihood	Impact	Risk
Spoofing	Attacker send the fake sensor data from room through guest wifi and fake devices	Middle	High	Middle
Tampering	Attacker modify the sensor data through guest wifi	Middle	High	Critical
Repudiation	Attacker prevent the sensor data through Man-in-the-Middle and prevent nothing happen	Middle	High	High
Info Disclosure	The attacker gained the flow sensor data through guest wifi since there is not encrypted communication. The information disposes of water flow usage and pattern.	High	Critical	High
Denial of Service	Attacker send a lot of traffic (sensor data) through guest wifi to disrupt the sensor system	Middle	High	Critical
Elevation of Privilege	Attacker got the staff credential through phishing email and website, then, they gain the sensor data through the staff account	High	Critical	High

Component 2: Web Dashboard (Same table structure)

Threat	Scenario	Likelihood	Impact	Risk
Spoofing	Attacker send the fake sensor data through guest wifi and fake devices to the web dashboard	middle	high	High
Tampering	Attacker modify the sensor data through guest wifi to web dashboard	middle	high	high
Repudiation	Attacker prevent the sensor data to dashboard through Man-in-the-Middle and prevent nothing happen during the peak time	high	Critical	high
Info Disclosure	Attackers gain the employee's credentials through phishing email, then they login to the dashboard and read all sensor data and guest information. They save it and keep it for further hijacking	high	Critical	high
Denial of Service	Attacker send a lot of traffic (sensor data) to the dashboard through guest wifi to disrupt the sensor system and dashboard	middle	Critical	high
Elevation of Privilege	Attackers gain the employee's credentials through phishing email and modify their authentication to authorize their account level and can modify the data on the dashboard.	high	Critical	high

Component 3: Cloud API (Same table structure)

Threat	Scenario	Likelihood	Impact	Risk
Spoofing	Attacker creates fake cloud endpoint; Sensor gateway sends all data to attacker's server	Middle	High	High
Tampering	Attackers modify the sensor data through guest wifi to the Cloud server. It reveals normal flow data, but it is essential overflow and may cause flooding	High	Critical	High
Repudiation	Attackers gain the sensor data during testing and save this data through cloud server due to encrypted communication, then the attacker can replay this data during down time. It caused the mess and further hijacking.	High	Critical	Critical
Info Disclosure	The attacker gained all guest information through the cloud server since they gained the credential through employees' phishing email.	High	Critical	high
Denial of Service	The attacker sends a lot of traffic through sensor gateway, and prevent the normal backup to cloud, it affects the normal backup data.	middle	high	high
Elevation of Privilege	The attacker gained the credential of the cloud through employees' credentials and modified their account to root and modify sensor data in the cloud server.	high	Critical	High

Component 4: Remote Controls (Gate/Shutoff) (Same table structure)

Threat	Scenario	Likelihood	Impact	Risk
Spoofing	Attacker creates fake huge sensor data through guest wifi and cause to shut off the device	Middle	High	Middle
Tampering	Attackers modify the sensor data through guest wifi due to encrypting communication, it reveals remote control is not secured.	Middle	High	High
Repudiation	Attackers gain the sensor data during testing and save this data through cloud server due to unencrypted communication, then the attacker can replay this data during down time. It caused the cutoff and further hijacking.	Middle	Critical	High
Info Disclosure	Attackers gain the sensor data and flow pattern, guest information, they can hijack the hotel and disclose the guest information to the public if they didn't pay for it.	High	Critical	Critical
Denial of Service	The attacker sends a lot of traffic through the sensor to the system during the peak time, and causes the hotel to shut off the device during the peak time, it causes the out of water in the guest room. It caused a huge loss in the hotel.	High	High	High
Elevation of Privilege	The attacker gained the high authentication since they steal employees' credentials and do the privilege of authentication and gain the access control of other systems. Gain the guest credit card information	High	Critical	Critical

For each scenario, be specific. "Attacker does bad thing" is not acceptable. Describe HOW they would execute the attack.

Section 5: Risk Summary

List your findings by risk level:

Critical Risks:

1. Unencrypted device communications (Info Disclosure) - Attackers can gain the sensor data and guest information through wifi, it is very dangerous since the attackers can realize the flow pattern, even they can gain the guest sensitive information through the employee's account or guest account since they are not unencrypted device. They can hijack using sensor data and guest information.
2. Unauthorized Access (Elevation of Privilege) is the most dangerous one since an attacker can control the whole system after they gained the control credential.
3. Repudiation - Attackers gain the sensor data during testing and save this data through cloud server due to unencrypted communication, then the attacker can replay this data during down time. It caused the cutoff and further hijacking.

High Risks:

1. Denial of Service - Attackers send a lot of traffic (sensor data) to the dashboard through guest wifi to disrupt the sensor system and dashboard during peak time, it may cause huge loss in the hotel.
2. Tampering - Attackers modify the sensor data through guest wifi due to encrypting communication, it reveals remote control is not secured.

Medium Risks:

1. Spoofing- Attacker send the fake sensor data from room through guest wifi and fake devices

Section 6: Recommended Mitigations

For each Critical and high-risk, propose a defense:

For Critical Risks:

Risk	Proposed Mitigation	Implementation Complexity
Weak dashboard authentication	Enable multi-factor authentication	Low – configuration change
Unauthorized Access (Elevation of Privilege)	Enable multi-factor authentication, biometric or token authentication	Low – configuration change
Unencrypted device communications (Repudiation)	Implement TLS encryption for all device-to-gateway communication	Medium – firmware update required

For High Risks:

Risk	Proposed Mitigation	Implementation Complexity
Denial of Service	Add rate limiting and anomaly detection for alert generation	Medium – cloud configuration
Dashboard tampering	Require admin approval for threshold changes; add confirmation dialogs	Low – application update
Spoofing	Add re-authentication requirement for bulk data exports	Low – application update